Topic Coverage and e-Assessment Topics Map for Biology for the IB MYP 4&5

	Chapter	Key concept	Related concept	Global context	Topic coverage (e-assessment topics in bold)
1	How is life organized?	Relationships	Patterns; Function	Identities and relationships	**Cells**: tissues, organs, systems, structure and function; Life processes; **Organisms:** interdependency, unity and diversity in life forms; classification
2	What chemical processes support life?	Systems	Transformation; Energy	Scientific and technical innovation	**Metabolism**: biochemistry and enzymes **Processes**: photosynthesis, cell respiration (aerobic and anaerobic), including word and chemical equations
3	How do organisms sustain themselves?	Systems	Energy; Environment; Balance	Personal and cultural expression	**Metabolism:** nutrition, digestion; gas exchange; circulation; transpiration and translocation**;** movement and transport, diffusion; osmosis **Organisms**: energy transfer and cycles [including nutrient, carbon, nitrogen]
4	What issues do larger organisms face?	Relationships	Form; Function; Development; Environment	Orientation in space and time	Physiology: transport systems, lungs, gut structure – i.e. issues relating to surface area: volume ratio
5	What factors affect human health?	Relationships	Consequences; Interaction	Fairness and development	**Cells:** factors affecting human health; vaccination
6	How do characteristics pass	Relationships	Evidence; Models;	Identities and	**Evolution**: DNA, inheritance, reproduction, life cycles

	from one generation to another?		Patterns	relationships	
7	How have different forms of life arisen?	Change	Interaction; Environment	Globalization and sustainability	**Evolution**: variation, natural selection; biodiversity; cell division, mitosis and meiosis; **Interactions with organisms**: competition, speciation and extinction
8	How are organisms adapted to survive?	Change	Environment; Interaction	Orientation in space and time	**Interactions with environment**: tropism, sense, nervous system, receptors and hormones
9	How do species interact?	Systems	Balance; Function	Globalization and sustainability	**Organisms:** habitat, ecosystem; interdependency, unity and diversity in life forms **Interactions with organisms**: pathogen/parasite, predator/prey, food chains/webs **Evolution:** biodiversity
10	How do the choices people make affect the environment?	Change	Environment; Balance	Globalization and sustainability	**Human interactions with environments:** human influences, habitat change or destruction, pollution/conservation; overexploitation, mitigation of adverse effects
11	How does biotechnology create new options in industry and health?	Change	Development; Transformation	Fairness and development	**Biotechnology:** genetic modification, cloning; ethical implications; genome mapping and application; 3D tissue and organ printing